

Week 1 Deliverable: Robotics Portfolio & Platform Comparison Report

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1. Executive Summary

This report provides a comprehensive architectural and regulatory analysis of the five public-facing robotics platforms within the inGen Dynamics portfolio: Fari, Senpai, Sentinel Prime AI, Aido Rover, and Aido Humanoid. The objective is to anchor engineering judgment to quantitative product constraints—such as power envelopes, compute classes, sensor stacks, and payload capabilities—using solely public domain documentation and reference architectures. This landscape view establishes the foundation for subsequent subsystem design, power tree analysis, and custom PCB fabrication.

2. Platform Comparison Matrix

The following matrix details the technical trade-offs and operational parameters across the five distinct robot classes:

Platform Name	Core Application & Target Use Case	Deployment Environment	Target Payload Capacity	Core Compute Class	Primary Sensor Stack Componentry	Regulatory Framework & Compliance Standards
Fari	Companion and eldercare assistant. Focuses on indoor environmental sensing, low-noise operation, and vitals monitoring.	Indoor	Low (~5–10 kg)	Low-power Edge MCU (e.g., ESP32-S3 / light NPU co-processor)	6-axis IMU, Time-of-Flight (ToF) distance sensors, ambient temperature/humidity sensors.	EU AI Act (Eldercare/Assistive classification), EN ISO 13482 (Safety requirements for personal care robots).
Senpai	Educational and conversational robot for learners. Prioritizes child-safe physical	Indoor	Minimal (<3 kg)	Application SoC (e.g., ARM Cortex-A series / Mid-	High-quality microphone array (I2S/PDM), low-light camera	EU AI Act Article 9 / Annex III (Educational emotional profiling

	envelopes and rich multimedia feedback.			range Mobile Processor)	module, basic ToF cliff sensors.	concerns), COPPA (Children's Online Privacy Protection Rule).
Sentinel Prime AI	Indoor security, patrol, and enterprise monitoring platform. Requires continuous real-time edge computer vision inference.	Indoor	Medium (~15–20 kg)	High-performance Edge AI Module (e.g., NVIDIA Jetson Orin NX / Hailo-8)	360° Vision Stack (MIPI-CSI / USB Cameras), 2D/3D LiDAR, Ultrasonic arrays, High-accuracy IMU.	EU AI Act Annex III (High-risk biometric/surveillance categorization), FCC Part 15 Subpart B (Unintentional Radiators).
Aido Rover	Outdoor patrol, perimeter security, and industrial site inspection. Features a ruggedized airframe and large battery runtime.	Outdoor	High (~30–50 kg)	Industrial/Automotive Grade Heterogeneous Compute (Dual Jetson Orin / High-TOPS SoC)	Long-range 3D LiDAR, Stereo Depth Cameras, RTK-GPS, GPS/GNSS, Infrared/Thermal Night Vision Cameras.	FCC Part 15 Subpart C (Intentional Radiators - LTE/Wi-Fi), CE-RED (Radio Equipment Directive), IP65/67 ingress protection.
Aido Humanoid	Bipedal research, manipulation, and advanced embodied AI demonstration platform.	Indoor / Semi-controlled Outdoor	High (Structural self-weight + active limb payload manipulation)	Distributed Heterogeneous Computing (High-TOPS VLM Inference Unit + Hard Real-Time Motor MCUs)	Full-body distributed IMU arrays, High-resolution joint encoders, Depth cameras, 6-axis tactile force/torque foot sensors.	IEC 61508 / ISO 13849 Functional Safety (PL/SIL compliance for active human-interactive actuators), EU AI Act.

3. Engineering Analysis & Design Trade-offs

3.1 Compute vs. Power Constraints

A stark architectural division exists between consumer-facing interactive platforms (**Fari**, **Senpai**) and high-bandwidth autonomous platforms (**Sentinel Prime AI**, **Aido Rover**).

1. **Passive Cooling and Low Acoustic Profiles:** For an eldercare assistant like **Fari**, fan noise introduces acoustic discomfort and compromises the user experience. Therefore, its compute hardware must stay within a strict thermal envelope ($<5\text{W}$), favoring distributed microcontrollers like the ESP32-S3 over full desktop-class application processors.
2. **Edge Inference Bandwidth:** Conversely, **Sentinel Prime AI** requires local multi-camera object detection (e.g., YOLOv8 pipelines at 15+ fps). This elevates compute needs into the 15W–45W range, necessitating dedicated switching regulators (Buck topologies) and active thermal management (forced convection or heavy heat sinking).

3.2 Sensor Topology & Protocol Implications

The choice of sensor communication protocol heavily affects signal integrity and layout complexity on the master PCB assembly:

- **Low-Speed I2C Bus Architectures:** Sensors utilized in the **Fari sensor hub** (IMU, ToF, OLED) communicate over a shared I2C bus. While routing is highly efficient (requiring only two lines: SDA and SCL), total bus capacitance (C_{bus}) must be maintained strictly under 400 pF to comply with Fast-mode specifications. To mitigate parasitic capacitance stemming from longer PCB traces, dedicated I2C buffers (e.g., PCA9517) must be placed to segment the bus.
- **High-Speed Vision Pipelines:** Sensor choices for **Sentinel Prime** and **Aido Rover** rely on MIPI-CSI, USB 3.0, or GMSL serialization to stream raw camera data. These high-speed differential pairs require tightly matched trace lengths, controlled differential impedances (90 Ohm or 100 Ohm), and isolated ground return paths to eliminate cross-talk and prevent electromagnetic interference (EMI).